

Complex Engineering Project (CEP)

CSE 345: Information System Design & Software Engineering

General Instructions: This is a Complex Engineering Project (CEP). The scenario below is deliberately open-ended so that no two students should produce identical designs. All submitted work must demonstrate individual analysis, judgment, and creativity; there is no single correct answer. Plagiarism in any form results in a zero mark for the entire submission. All UML diagrams must be produced using a proper CASE tool (draw.io, StarUML, PlantUML, Lucidchart, Visual Paradigm, or equivalent); hand-drawn diagrams receive zero marks for Task 3. Charts/schedules must be produced in a recognized project-management tool (MS Project, ProjectLibre, GanttProject, Trello, Notion, Monday.com, or an Excel Gantt template).

Course Outcomes & Learning Taxonomy

CO	Description	POs	Domain / Level
CO1	Discuss the software development life cycle and evolve methodologies for the design, analysis, planning and development of an information system.	PO1, PO2	Cognitive / L3
CO2	Use different project management, information gathering and requirements engineering principles on a S/W project.	PO2, PO3	Cognitive / L3, Affective / L2
CO3	Construct, analyze, combine and document UML and design patterns to work together in software design.	PO3	Psychomotor / L3, Affective / L2
CO4	Implement software testing and estimation techniques professionally for developing various categories of software subject to budget, schedule and platform constraints.	PO4, PO5	Cognitive / L3, Affective / L3

The Problem

MeghChain: Smart Cold-Chain Health Logistics Network

Digitizing Vaccine, Blood-Product, and Essential-Medicine Distribution Across a Disaster-Prone, Multi-Jurisdiction Hill Corridor Under Multi-Stakeholder, Multi-Constraint Conditions

The Sylhet–Meghalaya hill corridor of Bangladesh serves roughly 240 rural clinics, 9 district hospitals, and 3 border-crossing quarantine posts across 5 districts. Vaccines, blood products, and temperature-sensitive medicines are currently tracked using paper cold-chain logs, radio call-ins, and informal courier networks. Following the flash floods of March 2026, which submerged two of the corridor's three primary access roads for eleven days, at least 40,000 vaccine doses were confirmed spoiled due to undetected cold-chain breaks, and the region's fragility as a health-logistics corridor became a national emergency.

A newly formed inter-agency body, MeghChain Health Logistics Trust, has been mandated by the Ministry of Health and Family Welfare and co-funded by a regional development bank (USD 65 million, disbursed in tranches contingent on independently audited delivery milestones) to digitize cold-chain visibility and distribution coordination across the corridor. The Trust brings together the Directorate General of Health Services (DGHS), the Expanded Programme on Immunization (EPI) unit, two private pharmaceutical distributors, a regional Red Crescent chapter, and district civil-surgeon offices.

The problems the Trust faces are not purely technical, they span engineering, public health policy, human behavior, and cross-border regulation.

Constraints

- Six of the corridor's clinics have only intermittent mobile network coverage, and two have none for days at a time during monsoon season.
- A 2011-era EPI vaccine-stock reporting system cannot be decommissioned before the end of Phase 1 and must continue to receive data.
- Customs clearance at the border quarantine posts is handled by a separate authority (NBR) running a disconnected legacy platform, and cross-border data sharing is politically sensitive.
- Private distributors are reluctant to expose real-time stock and routing data to competitors or to the Trust itself, citing commercial confidentiality.
- Community health workers in two districts previously abandoned a smart-thermometer pilot after rumors spread that the devices were used for surveillance; trust must be rebuilt.
- Blood-product handling is subject to stricter chain-of-custody and consent regulations than general medicine, adding a distinct compliance track.
- Phase 1 must be delivered within 12 months on a budget of USD 6.2 million, and the platform must remain auditable to the funding body at every milestone.

MeghChain's leadership requires a rigorous, evidence-based software engineering approach that identifies and structures the operational problems, proposes and justifies a viable design and implementation strategy, and demonstrates awareness of the technical, human, regulatory, and public-health dimensions of the challenge. You are expected to critically analyze the problem, evaluate trade-offs, and defend your chosen approach; generic or template-style answers will not receive full marks.

Assignment Tasks

The following four tasks must each be addressed as clearly separated, titled sections in your report. All design decisions must be explicitly traceable to specific details of the MeghChain scenario.

#	Task Description	CO	Marks
1	Investigate the MeghChain scenario and identify the operational, technical, organizational, and human factors that make this an ill-structured problem. Propose and justify a software development approach (e.g., a specific lifecycle model or a hybrid/tailored methodology) that you believe is best suited to the constraints described. Your justification must explicitly weigh at least two alternative approaches you rejected, and explain, with reference to specific details of the scenario (connectivity, phased rollout, stakeholder distrust, disaster context, etc.), why your chosen approach is more defensible than the alternatives.	CO1	4
2	Elicit and structure the requirements of the MeghChain platform. Distinguish between functional, non-functional, and constraint-driven requirements, and reflect the perspectives of at least four stakeholder groups with conflicting interests. Propose a prioritization framework of your own design (not simply MoSCoW recited without adaptation), apply it, and discuss at least two situations where satisfying one stakeholder's requirement actively works against another's. In addition, produce a high-level project schedule (Gantt chart or WBS-based schedule) for Phase 1, showing how your requirement priorities translate into a sequencing and resourcing strategy under the given timeline and budget.	CO2	5

#	Task Description	CO	Marks
3	Design the MeghChain system using UML. You must include at least four distinct UML diagram types (structural and behavioral) that together communicate the complexity of the problem — choose the set that best fits what you are trying to show, not simply the four easiest to draw. Apply at least two recognizable design patterns (e.g., for offline-first synchronization, role-based access, event handling, or integration with legacy systems) and justify why each pattern fits a specific constraint in the scenario. Every diagram must be accompanied by written commentary explaining the design decisions made, the alternatives you considered and rejected, and which specific detail of the MeghChain scenario drove the decision.	CO3	7
4	Propose a testing and quality-assurance strategy appropriate to the categories of software involved in MeghChain (e.g., offline-capable field applications, integration middleware, a central dashboard), and justify your choice of testing levels and techniques given the budget, schedule, and platform constraints. Then produce a defensible effort/cost estimate for Phase 1 using a named estimation technique (e.g., Function Point Analysis, COCOMO II, or Planning Poker/story-point-based estimation), showing your inputs and assumptions. Critically discuss at least two sources of estimation risk specific to this project and how your plan accounts for them.	CO4	4
	Total		20

Assessment Rubric

Task	Excellent (90–100%)	Satisfactory (60–89%)	Needs Improvement (<60%)
Task 1 Requirements & Methodology	Approach is explicitly justified against ≥ 2 rejected alternatives; prioritization framework is original and consistently applied; stakeholder conflicts are analyzed with concrete trade-offs.	Approach and prioritization are present and reasonable but justification is generic or only lightly tied to the scenario; conflicts are named but not deeply analyzed.	Methodology or prioritization is asserted without justification; stakeholder analysis is superficial or missing; requirements are generic/template-like.
Task 2 Requirements, Scheduling & Stakeholders	≥ 4 stakeholder perspectives clearly reflected; schedule logically sequences work against real constraints; trade-offs between stakeholders are explicit and well-argued.	Requirements and schedule are reasonable and mostly traceable to the scenario but some stakeholder perspectives or trade-offs are thin.	Requirements are not clearly categorized; schedule is generic or not connected to stated constraints; stakeholder conflicts are ignored.
Task 3 UML Design	≥ 4 well-chosen diagram types with strong internal consistency; ≥ 2 design patterns correctly applied and justified; commentary clearly ties each decision to a specific scenario detail.	Required diagrams present and mostly consistent; patterns present but justification is thin; commentary is present but partly generic.	Fewer than 4 diagram types, diagrams inconsistent with each other, patterns missing/misapplied, or little to no analytical commentary.

Task	Excellent (90–100%)	Satisfactory (60–89%)	Needs Improvement (<60%)
Task 4 Testing & Estimation	Testing strategy is matched to distinct software categories in the system; estimation technique correctly applied with clear assumptions; ≥ 2 real risks identified with mitigations.	Testing and estimation approaches are reasonable but generic; assumptions are present but not well justified; risks are named but shallow.	Testing strategy is generic/one-size-fits-all; estimation technique missing or misapplied; no meaningful risk discussion.

Submission Guidelines

- Submit a single PDF document with a cover page (name, ID, section, date), a table of contents with page numbers, and clearly numbered sections for each task.
- UML diagrams must be produced with a proper CASE tool (draw.io, StarUML, Lucidchart, PlantUML, Visual Paradigm, or equivalent). Hand-drawn diagrams receive zero marks for Task 3.
- The Phase 1 schedule must be produced in a recognized project-management tool (MS Project, ProjectLibre, GanttProject, Trello, Notion, Monday.com, or an Excel Gantt template); screenshots must be high resolution.
- All figures, tables, and diagrams must be numbered and captioned (e.g., “Figure 3: Sequence Diagram Cold-Chain Breach Alert v1.0”).
- Maximum report length: 40 pages, excluding cover page, table of contents, and references. There is no minimum, quality over quantity.
- Late submissions are penalized 10% per day. No extensions will be granted unless approved in writing by the course instructor at least 48 hours in advance.